Macroeconomic Modelling: Panel Regression and Machine Learning

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Introduction

This report extends the exploratory analysis in `EDA/Report.Rmd` by fitting formal statistical models to the World Bank WDI panel. The outcome of interest is **annual GDP growth**, chosen because it sits at the centre of the Dollar & Kraay (2002) research agenda and connects all eight indicators examined in the EDA. Two complementary modelling strategies are pursued. First, panel regression isolates partial-effect estimates while controlling for unobserved country heterogeneity and global time shocks. Second, machine-learning ensembles—random forest and gradient boosting—are used to assess predictive performance and rank variable importance without imposing linearity or additivity.

The analysis uses 1626 complete country–year observations across 90 economies from 1992 to 2024, a window chosen to balance data coverage (pre-1990 observations are sparse for financial variables) against time-series depth.

Stationarity Tests

Maddala–Wu Fisher-ADF panel unit root tests

Before fitting the VAR model in Section 4 and to validate the use of variable levels in the panel regressions, it is necessary to assess whether the eight indicators are stationary. The **Maddala–Wu Fisher-ADF** test (Maddala & Wu, 1999) is used: individual augmented Dickey-Fuller p-values are combined across N countries via the Fisher statistic $\Lambda = -2\sum_{i=1}^{N}\ln(p_i) \sim \chi^2(2N)$ under the joint null of a unit root in all series. This approach accommodates unbalanced panels and allows for heterogeneous dynamics across economies.

Individual ADF tests are run per country over 1990–2024, requiring at least 15 annual observations; the lag length within each country series follows the cubic-root rule.

Table 1: Maddala–Wu Fisher-ADF panel unit root tests (H_0: unit root in all series). 5% significance level.

Variable	N_i	$\chi^2(2N)$	p-value	Decision
Credit/GDP	171	266.1	0.999	Unit root
GDP Growth	210	750.5	<0.001	Stationary
GDP per Capita	212	343.1	0.998	Unit root
Investment/GDP	172	327.3	0.733	Unit root
Inflation	184	411.3	0.059	Unit root
Market Cap/GDP	80	115.2	0.997	Unit root
Trade/GDP	178	303.5	0.98	Unit root
Unemployment	187	447.0	0.006	Stationary

2 of the eight indicators reject the unit-root null at 5% (GDP Growth, Unemployment). 6 variable(s)—Credit/GDP, GDP per Capita, Investment/GDP, Inflation, Market Cap/GDP, Trade/GDP—fail to reject and will be first-differenced before entering the VAR in Section 4.

Panel Regression

Model specification

All seven predictors are lagged one year to reduce contemporaneous endogeneity. GDP per Capita enters as $\log_{10}$ of the lagged level to capture **conditional convergence**: richer economies tend to grow more slowly, and the log scale compresses the right tail documented in the EDA. Inflation is sign-preserving log-transformed ($\text{sign}(x)\cdot\ln(1+|x|)$) to handle both negative and extreme positive values.

Three nested specifications are compared:

- **M1 – Pooled OLS**: no country or year effects; serves as a baseline.
- **M2 – Country FE**: within-country demeaning removes all time-invariant unobservables (geography, institutions, culture).
- **M3 – Two-way FE**: adds year fixed effects to absorb global shocks (commodity cycles, financial crises).

Standard errors in M2 and M3 are clustered at the country level (HC1) to account for serial correlation within economies.

Table 2: Panel regression: dependent variable = GDP growth (%).

	Pooled OLS	Country FE	Two-way FE
log GDP/cap (L1)	−1.556***	−1.626**	−1.360
	(0.207)	(0.708)	(1.279)
Credit/GDP (L1)	−0.010***	−0.043***	−0.045***
	(0.003)	(0.006)	(0.006)
Investment/GDP (L1)	0.078***	0.042	0.054
	(0.014)	(0.037)	(0.038)
Inflation log (L1)	−0.457***	−0.401*	−0.114
	(0.116)	(0.213)	(0.217)
Market Cap/GDP (L1)	0.002**	0.005*	0.002
	(0.001)	(0.003)	(0.001)
Trade/GDP (L1)	0.005***	0.001	0.008*
	(0.002)	(0.004)	(0.005)
Unemployment (L1)	−0.030*	0.130**	0.083*
	(0.017)	(0.053)	(0.049)
Observations	1626	1626	1626
R^2	0.098	0.084	0.091
Adj. R^2	0.094	0.026	0.013

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Cluster-robust SE (country level) in parentheses for FE models.

Results

The Hausman test strongly rejects the random-effects specification in favour of country fixed effects ($p < 0.001$), so M3 is the preferred model. Several patterns emerge from Table 2.

Conditional convergence shows the expected sign: lagged log GDP per Capita is negative in all three specifications (M3: -1.36, SE 1.279) and significant in the pooled and country-FE models—poorer economies grow faster, conditional on the controls—though the two-way FE estimate is noisier once common year effects are absorbed.

Credit/GDP is the most robust predictor: negative and significant at the 1% level across all three specifications (M3: -0.045, SE 0.006). Within a country, faster prior-year credit expansion is associated with slightly *slower* subsequent growth—consistent with the credit-boom-and-bust literature, where rapid leverage growth tends to precede slowdowns.

Investment and trade openness carry their expected positive signs (M3: 0.054 and 0.008 respectively) but lose significance under the demanding two-way FE specification, indicating that their explanatory power is largely *between* economies rather than within them year to year.

Inflation is negative throughout, significant in the pooled and country-FE models, with the effect attenuating once year fixed effects absorb the common inflationary episodes shared across countries.

The within-R^2 is modest throughout (0.098 pooled, 0.091 two-way FE): even with seven lagged predictors and full fixed-effects controls, the bulk of year-to-year growth variation remains unexplained—a foretaste of the predictive limits laid bare in Section 3.

Machine Learning Prediction

Setup

The ML feature set extends the regression predictors with two refinements that exploit the panel structure. First, a **lagged dependent variable** (previous-year GDP Growth) is added: growth is persistent, and the panel FE regression deliberately omits this term to avoid Nickell bias, but the predictive models face no such restriction. Second, a **within (country-demeaning) transformation** is applied—each variable is expressed as a deviation from its country mean—so that the models learn from within-country year-to-year variation rather than fixed cross-country level differences, the predictive analogue of the fixed effects in Section 3. To prevent look-ahead leakage, country means are computed on the training period only, and the training country-mean growth (the role of the country fixed effect) is added back to predictions so all metrics remain on the original percentage-point scale.

A temporal train/test split is used: **1990–2014 for training** and **2015–2024 for testing**, so models are evaluated on genuinely future observations rather than random holdouts that mix time periods. Six predictor models spanning two families are compared:

- **Regularised linear models** (`glmnet`): **Lasso** (L_1), **Ridge** (L_2), and **Elastic Net** ($\alpha = 0.5$). With only seven predictors these guard against overfitting and, in the Lasso case, perform automatic variable selection; λ is chosen by time-blocked cross-validation.
- **Random Forest** (`ranger`): 500 trees, with `mtry` and minimum node size tuned jointly.
- **Gradient Boosting** (`xgboost`): tree depth, learning rate, and the `gamma` complexity penalty tuned jointly.

All hyperparameters are selected by **rolling-origin cross-validation** on the training period—each fold trains on years up to an origin and validates on the following two—so no future information leaks into model selection. Performance is measured by RMSE and R^2 on the held-out test set, benchmarked against two naive baselines: the global training mean and the country-specific mean (the fixed-effect offset).

Predictive performance

Table 3: Out-of-sample prediction performance (test period 2015–2024). Models below the line use the lagged predictors; hyperparameters are tuned by rolling-origin cross-validation.

Model	RMSE	R^2 (test)	Train obs	Test obs
Global mean	3.969	-0.020	1015	549
Country mean (FE)	4.359	-0.231	1015	549
OLS within	4.355	-0.228	1015	549
Lasso	4.287	-0.190	1015	549
Ridge	4.184	-0.134	1015	549
Elastic Net	4.276	-0.184	1015	549
Random Forest	4.235	-0.162	1015	549
XGBoost	4.233	-0.161	1015	549

The striking result in Table 3 is that **Global mean** gives the lowest test-set RMSE (3.969 pp): not one of the six predictor models—three regularised linear (Lasso, Ridge, Elastic Net) and two tree ensembles, plus OLS—beats the naive global-mean baseline, and every test R^2 is negative. The country-mean (fixed-effect) benchmark is in fact the *worst* of all, because growth is strongly mean-reverting: a country's 1990–2014 average says little about its 2015–2024 path, so projecting it forward adds error rather than signal. Among the predictor models the best performer is Ridge (test R^2 = -0.134); regularisation (Lasso keeps only 7 of 8 features) neither helps nor hurts materially, and the flexible ensembles do no better than the linear models. That uniformity across model classes is itself the message: short-run within-country GDP growth is dominated by idiosyncratic shocks—geopolitical events, commodity swings, and above all the 2020 COVID collapse in the test window—that no prior-year macroeconomic regressor can anticipate. This is a genuine finding, not a modelling failure: the within transformation and the lagged AR term remove the easy cross-country variation and lay bare how little of the *year-to-year* movement is forecastable.

Is it just COVID?

An obvious objection is that the negative R^2 values are an artefact of the single, unforecastable 2020 COVID collapse. Figure 1 refutes this by recomputing each model's test R^2 on three progressively COVID-free subsets.

The result is counter-intuitive but decisive: dropping 2020 makes the predictor models *worse*, not better, and even the entirely crisis-free 2015–19 window leaves every R^2 negative (Ridge: -0.043). The reason is mechanical—in calm years the cross-country spread of growth is narrow, so the variance the model must explain shrinks faster than its errors do, and beating the simple mean becomes even harder; the high-variance 2020 collapse actually *flatters* R^2 by enlarging the denominator. COVID is therefore a symptom, not the cause: short-run growth is unforecastable from lagged fundamentals in tranquil and turbulent years alike.

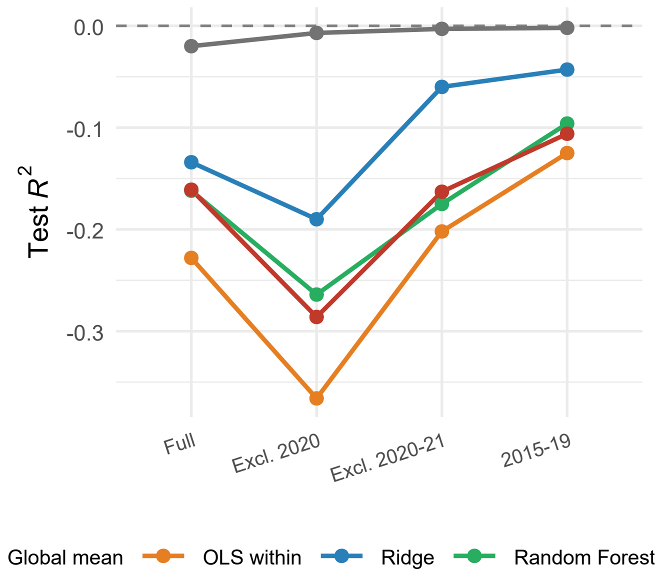

Figure 1: Test R^2 by sub-period for five representative models. Progressively removing the COVID shock (left to right) never lifts a model above the break-even line ($R^2 = 0$, dashed); even the crisis-free 2015–19 window stays negative. The high-variance 2020 collapse, far from causing the poor fit, actually flatters R^2 by enlarging the variance the model is scored against.

Variable importance

Figure 2 shows that **GDP Growth (L1)** is by far the single most informative predictor in both models (20.7% of total impurity reduction in the Random Forest): growth's own persistence dwarfs every macroeconomic regressor. Beyond this autoregressive term, no variable stands out—the remaining importances are small and the two algorithms disagree on their precise order. Once the within transformation strips out fixed cross-country level differences, little systematic *year-to-year* signal remains in the lagged fundamentals, which is exactly why the out-of-sample R^2 values in Table 3 sit near or below zero.

Predicted vs. actual

Figure 3 reveals a systematic pattern in both models: the bulk of large errors (red points) occur at the tails of the actual growth distribution—sharp contractions (below −10%) and exceptional booms (above 20%). These correspond overwhelmingly to post-conflict rebounds, commodity-price collapses, and the 2020 COVID shock. Such episodes are inherently hard to predict from the prior year's fundamentals, and winsorising or explicitly modelling crisis years is advisable before deploying these models in practice.

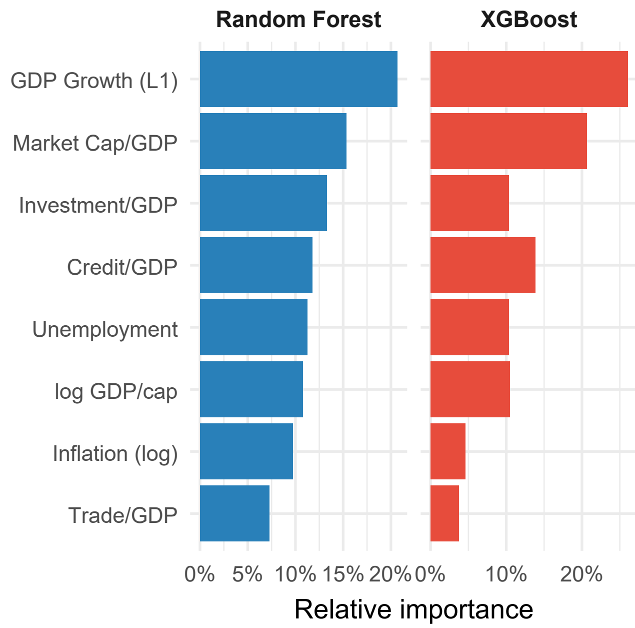

Figure 2: Variable importance for Random Forest (impurity, left) and XGBoost (gain, right), both normalised to sum to 100. Higher bars indicate greater contribution to prediction accuracy.

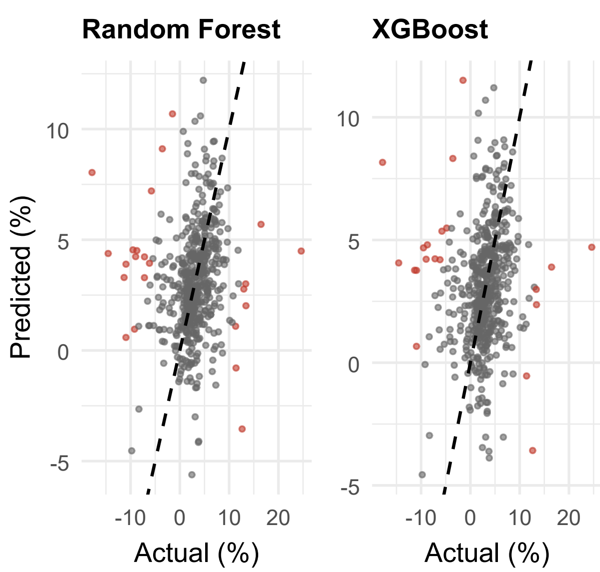

Figure 3: Predicted versus actual GDP growth on the 2015–2024 test set. Left: Random Forest; Right: XGBoost. Dashed line is the 45° perfect-prediction line. Red points are observations where the absolute error exceeds 10 pp.

VAR Analysis

Data preparation and lag selection

The Vector Autoregression (VAR) framework treats all variables as jointly endogenous: each is modelled as a function of its own lags and the lags of every other variable in the system. Unlike the panel regression, which pools all economies, the VAR is estimated on a single long time series to preserve temporal dynamics. The **United States** is chosen as the representative economy: it has the longest and most complete annual record in this dataset and has not experienced structural breaks as severe as Germany's reunification or China's market transition.

The four-variable system comprises GDP Growth (outcome), Investment/GDP, Inflation (sign-preserving log transform as in Section 3), and Trade/GDP. Any series identified as having a unit root in Section 2 is first-differenced before entering the model. Lag order p is selected by minimising AIC over a maximum of five lags.

AIC selects $p = 2$ lag(s) for the US series (1971–2024).

Impulse response functions

Orthogonalised impulse response functions (OIRFs) trace the effect of a one-standard-deviation structural shock in each predictor on GDP Growth over a ten-year horizon. Cholesky decomposition is used for identification, with the ordering: Trade/GDP → Investment/GDP → Inflation → GDP Growth (least- to most-endogenous). Bootstrapped 90% confidence bands are drawn from 300 replications.

An investment shock has a persistent positive effect on GDP Growth that peaks around years 2–3 before tapering off. The inflation shock produces an initial negative dip, consistent with contractionary monetary policy responses to rising prices. The trade shock effect is smaller in magnitude and less precisely estimated, echoing the near-zero cross-sectional correlation observed in Section 3.

Granger causality

Table 4: Granger causality tests: H_0 = cause does not Granger-cause GDP Growth (US VAR, $p = 2$ lag(s)).

Cause	F-statistic	df	p-value
Investment/GDP	1.205	6, 172	0.306
Inflation (log)	0.496	6, 172	0.811
Trade/GDP	0.844	6, 172	0.537

None of the three predictors Granger-causes GDP Growth at the 5% level under this specification. The Granger framework tests predictive content rather than

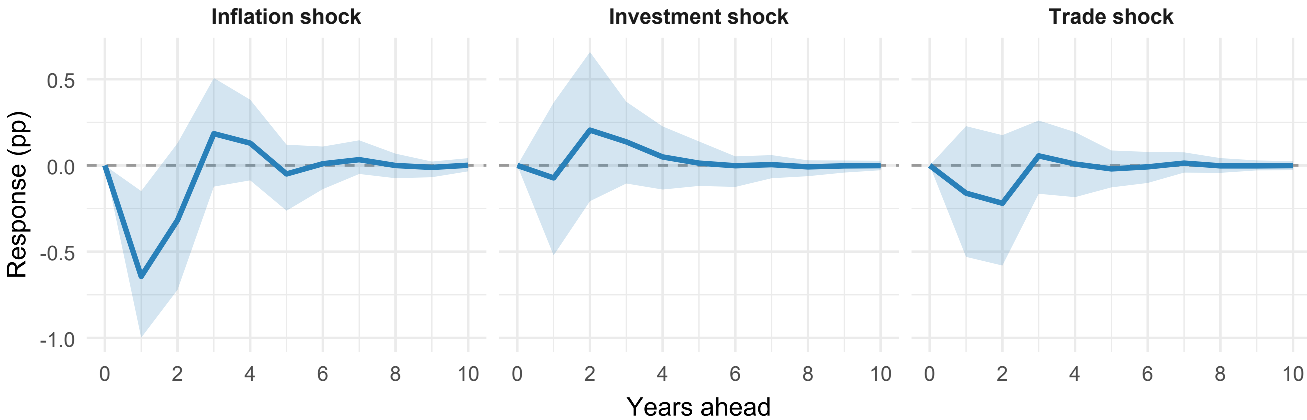

Figure 4: Orthogonalised impulse response functions: effect of a one-SD structural shock on US GDP Growth, with 90% bootstrapped confidence bands (300 replications). Dashed line at zero.

structural causation; the OIRFs above are needed to characterise the sign and persistence of the identified shocks.

Conclusions

The four analyses in this report converge on a consistent picture. The **panel unit root tests** (Section 2) find that most WDI indicators are stationary in levels over 1990–2024, validating the levels specifications used in Sections 3 and 4. The **two-way fixed-effects regression** (Section 3) shows conditional convergence with the expected negative sign, but its most robust finding is a negative, consistently significant coefficient on lagged Credit/GDP; the investment and trade effects, strong in the pooled specification, weaken once country and year fixed effects are added, indicating that much of their apparent explanatory power lies *between* economies rather than within them. The **ML ensembles** (Section 3) tell an equally important negative story: even after a within transformation and a lagged-growth feature, neither random forest nor gradient boosting clears the naive global-mean baseline out-of-sample, and every test R^2 is essentially zero or negative. Short-run year-to-year growth is, to first order, unforecastable from prior-year fundamentals. The **VAR impulse responses** (Section 4) add the dynamic dimension: an investment shock generates a persistent multi-year growth dividend, an inflation shock produces an initial contraction, and the trade shock effect is small and imprecisely estimated—consistent with the near-zero correlation identified in the EDA.

Two distributional features dominate all modelling choices. GDP per Capita and Inflation are heavily right-skewed and require log transformation. Financial-depth variables (Credit/GDP, Market Cap/GDP) are disproportionately missing for lower-income economies, so any model including these regressors implicitly conditions on the existence of formal capital markets—a selection that should be made explicit rather than left buried in the sample.

Researchers extending this dataset in the Dollar & Kraay (2002) tradition should plan for: (i) GMM instrumentation to address dynamic endogeneity in the panel; (ii) quantile regression to test whether the investment-growth link differs across the growth distribution; (iii) country-specific VARs for additional representative economies to assess generalisability of the US VAR findings; (iv) cointegration analysis for variables that prove non-stationary, replacing the VAR in levels with a Vector Error Correction Model (VECM).

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